BEST Torino | NTT DATA

Hackathon Challenge 2026

Case Study Challenge

Applying Innovative AI Technologies (ML/AI/GenAI/AgenticAI) to R&D Engineering and Product Development Processes.

Saturday - 28th March 2026
08:30 - 18:30

Politecnico di Torino
Turin, Italy

Direct contact with 4 Process Expert Consultants
NTT DATA Italia

Innovate the Future of Engineering

Analyze. Cluster. Prioritize.
Make vs Buy.

Event at a Glance

Target Audience

143 Participants

Selected among all Engineering faculties at PoliTo.

Team Structure

Groups of 4 Students

Collaborative working groups formed to tackle the Case Study challenge.

Expert Mentorship

4 NTT DATA Technical Mentors

1 Technical Mentor per classroom during all working hours for guidance and support.

Language

All deliverables and final presentation must be in **English**.

Format

- On-site Hackathon
- 08:30 - 18:30
- Politecnico di Torino

Event Day Schedule - Saturday 28th March 2026

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08:30
Welcome
Check-in & Networking

09:00
Official Opening
BEST PMs, PoliTo Prof. Molinari, NTT DATA Presentation

09:40
Challenge Briefing
Topic Explanation

10:15
Hackathon Start
Start of Working

11:30
Coffee Break
Recharge

13:30
Lunch
Buffet

14:30
Work Session
Deep Focus

18:00
Submission
Submission Deadline

18:15
Closing
Final Aperitif

One NTT DATA delegate will be present in each classroom for support throughout the day.

AI Landscape: Definitions & Concepts

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Evolution of Intelligence

Artificial Intelligence (AI)

Broad field of systems performing tasks that typically require human intelligence (e.g. reasoning, learning, problem-solving, etc.).

Machine Learning (ML)

Subset of AI. **Data-driven algorithms** that learn patterns from historical data to make predictions or decisions without explicit programming.

Generative AI (GenAI)

Subset of ML using Deep Learning models (LLMs) to generate **new content** (e.g. text, code, images, etc.) based on learned patterns.

Agentic AI

Goal-driven agents that can **plan**, **reason**, **use tools** (e.g. APIs, web search, etc.) and **act autonomously** with feedback loops to achieve complex objectives.

Goal

Create a **Hybrid Workforce (Engineer + AI)** where AI handles repetitive tasks allowing engineers to focus on high value-added tasks and innovation.

Technical Glossary

Model & LLM (Large Language Model)

A **Model** is the mathematical "engine" learned from data. An **LLM** is a specific type of model trained on vast text data to understand and generate human language (e.g. GPT-5, Llama 3, etc.).

Dataset

The structured collection of **data used to train or fine-tune a model**.

Inference

The process of **running a trained model** to make predictions or generate outputs on new data.

Prompt

The **input text or instruction given to a GenAI model** to elicit a specific response.

RAG (Retrieval-Augmented Generation)

A **technique** that enhances model accuracy by **retrieving relevant facts from external verified sources** (e.g. company databases, set of input documents, etc.) before generating an answer reducing hallucinations.

Hallucination

When a model generates incorrect, nonsensical or "made-up" information confidently.

Zero-shot / Few-shot

The ability of a model to perform a task with no examples (zero-shot) or very few examples (few-shot).

The Hybrid Workforce Objective

Strategic Purpose

- **Automate repetitive and low value-added tasks** to free up engineering time.
- **Augment human capabilities** on complex and high-impact innovation work.
- **Shift focus** from "doing" routine work to "designing" outcomes.

Key Benefits

- **Higher Productivity:** Faster development cycles and reduced time-to-market.
- **Improved Quality:** Consistency in outputs and reduction of human error.
- **Knowledge Reuse:** Leveraging historical data to solve new problems efficiently.

Core Principles

- **Human-in-the-Loop:** Engineers validate critical AI outputs.
- **Transparency:** AI decisions should be explainable and traceable.
- **Responsible AI:** Security, compliance and ethics by design.

Products as Systems: Sensors-Logic-Actuators

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Complex Engineered Products

In structured engineering sectors a product is often a complex system composed of three core functional blocks that continuously interact between each other:

Sensing (Inputs)

Sensors collect data from the physical environment (e.g. temperature, pressure, speed, vision of environmental state and situation, etc.) to monitor the system's state.

Control Logic (Processing)

Microcontrollers (ECUs) process input data using control algorithms to make decisions and calculate necessary adjustments and commands to be sent to actuators.

Actuation (Outputs)

Actuators (e.g. motors, valves, heaters, displays, etc.) execute the control decisions physically to change the system's state basing on received commands from microcontrollers (ECUs).

Multidisciplinary Collaboration:

Hardware (HW) | Software (SW) | Mechanics (MEC)

Industry Variations

While complex engineered products fit the sensors-logic-actuators model, other industries produce different types of products that could still benefit from AI application in their **production processes**, **supply chain processes** and **sales management processes**.

Chemical Nature

Food, Beverages, Oils, Pharmaceutical Products, etc.

Focus: Recipe Optimization, Quality Control, Customer Relationship Management, Demand & Forecasting and Trade Promotion Management.

Purely Mechanical

Bearings, Fasteners, 3D-Printed Components, etc.

Focus: Material Selection, Quality Control and Stress Analysis.

Key Takeaway

Even simple products have complex manufacturing processes, supply chain processes and sales management processes - perfect for AI/ML optimization.

Core Definitions

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Process

A repeatable set of activities that transforms inputs into outputs to achieve specific objectives. It defines ***what*** needs to be done.

Methodology

The structured approach, specific practices, rules and guidelines applied to execute a process effectively. It defines ***how*** it is done.

Tool

The **enabling software**, **platform**, or **instrument** that supports the execution of process steps and methodology implementation.

KPIs (Key Performance Indicators)

Quantitative metrics to measure process efficiency/effectiveness.
Crucial for continuous process improvement.

Process Standard

International norms defining mandatory process contents (***what***) for specific industries leaving implementation (***how***) to company-specific choices and decisions.

Industrial Context

Understanding core definitions is essential for applying AI effectively within structured engineering environments.

V-Model Overview

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1: Market Analysis (e.g. Customer feedbacks, user groups, generation of ideas/scenarios, state of the art monitoring with respect to regulations, processes and technologies, etc.)

2: Requirements Engineering (e.g. requirements elicitation, requirements analysis, comparing and summarizing documents, deriving requirements from regulations, requirements classification, requirements prioritization, etc.)

3: Architectural Design (e.g. architectural design specifications, model generation, interface design, creation of sequence diagrams, text-to-model, model-to-text, visualization of dependencies, etc.)

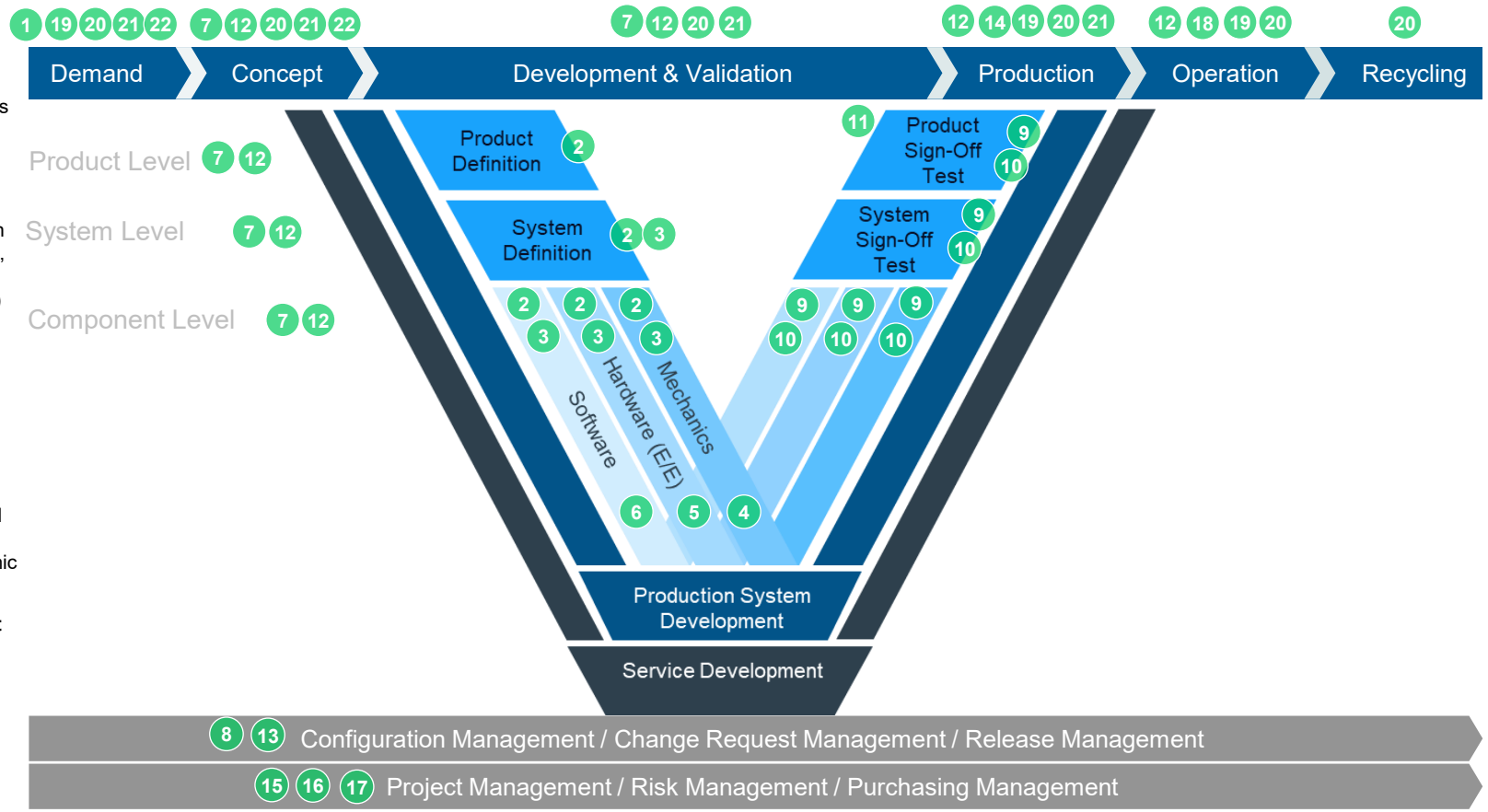
4: Assistants for Mechanical Design (e.g. assistants for CAD/CAE/CFD, automatic model generation and changes, model change analysis, model similarity search, automatic generation of design parameters, etc.)

5: Assistants for Hardware Design (e.g. assistants for electronic HW design, PCB layout optimization, automatic schematic generation and changes, schematic change analysis, schematic similarity search, automatic generation of electronic parameters for HW simulations, etc.).

6: Assistants for SW Design and Development (e.g. automatic creation or optimization of source code, automatic code reviews, automatic code quality checks against MISRA rules, automatic code documentation and traceability, etc.)

7: Quality Analysis (e.g. automatic checks for traceability and consistency, checks for accuracy, standardization potentials analysis, analysis of dependencies, prediction of error sources, evaluation of design alternatives, formal reviews of processes and Work Products based on checklists, etc.)

8: Change Request Management (e.g. identification and tracking CRs, dependency analysis between CRs, CR impact analysis, CR effort and cost estimation, Change Control Board decisions, CR approval, CR implementation planning, CR validation and tracking to closure, bidirectional traceability between CRs and impacted Work Products, etc.)



15: Project Management (e.g. project planning, automatic generation of project work breakdown structure, effort estimation, resource allocation, periodic schedule adjustment and optimization, interface definition with communication plan, task prioritization, project progress reports, lessons learned, etc.)

16: Risk Management (e.g. identification and tracking of risks, classification and assessment of risks, risk prioritization, risk root cause analysis, risk impact analysis, definition of risk mitigation actions, risk monitoring, corrective actions, etc.)

17: Purchasing Management (e.g. management of Purchase Requisitions (PRs), Supplier Qualification, Drafting of RFQ document for Suppliers, Tender Management, Comparative Analysis of Supplier offers, Supplier Selection, smart negotiation with Suppliers, emission of Purchase Orders (POs), management of Good Receipts (GRs), emission of invoice authorizations, registration and payment of invoices, etc.)

18: Technical Manual Management (e.g. generation of user manuals and repair manuals based on engineering technical specifications, translation of user manuals and repair manuals in multiple languages, etc.)

19: Sales Management (e.g. Demand and Forecasting, Trade Promotion Management and Optimization with ROI analysis, process optimization within CRM tools, marketing campaigns, monitoring of market trends, etc.)

20: Cost Engineering (e.g. cost analysis and optimization, analysis of cost historical data, competitive analysis, cost-benefit analysis, etc.)

21: Variant Management (e.g. platform solution design, optimization of product variants, feature identification and mapping with product variants, etc.)

22: Innovation Management (e.g. startup scouting, idea validation, patent management, patent search and classification, etc.)

9: Test Definition and Planning (e.g. test environment definition, selection of test methods, generation of test cases from requirements and architecture keeping bidirectional traceability & consistency, test steps with related acceptance criteria, test planning with smart selection of test cases to be executed for each release, optimization of non-regression tests for each release, etc.)

10: Test Data Management (e.g. test input data preparation, automated test execution, test results collection with proper evidence, automatic defect identification, automatic test report generation, synthetic summary of test loops, testing coverage metrics, bidirectional traceability between test results and test specifications, etc.)

11: Product Homologation (e.g. product data management, identification of potential violations of regulatory requirements, creation of homologation documents and evidence, smart transmission of homologation documents to authorities, etc.)

12: Issue Resolution Management (e.g. identification and tracking of issues, classification of issues, issue root cause analysis, issue impact analysis, issue solution identification, issue resolution effort and cost estimation, approval of issue resolution, issue resolution planning, issue resolution validation, issue tracking to closure, traceability between issues and impacted Work Products, etc.)

13: Configuration Management (e.g. clear identification of configuration items, definition of configuration management strategy, smart baselining strategy, release planning, control of modifications of each configuration item, automatic checks on naming conventions, smart backup and restore strategies, etc.)

14: Production Planning (e.g. production plant layouts and machineries, production task list, production safety checklists, work order management, management of supply chain for production raw materials, etc.)

R&D and Product Development Processes

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Requirements Engineering

- **Elicitation:** Business Needs, Customer Requirements, Market Analysis.
- **Analysis:** Functional Requirements, Performance Requirements, Environmental Requirements.
- **Management:** Links for Bidirectional Traceability and Consistency between different specifications, Change Requests Management, Change Control.

System & SW Design

- **System:** Architecture, Allocation of SYS requirements to HW/SW/MEC components.
- **Software:** Architecture, Detailed Design, Coding.

Verification & Validation

- **SW Verification and Integration:** SW Unit Tests, SW Integration Tests, SW Qualification Tests and SW builds for each stable release.
- **System Integration:** Component & System build.
- **Validation:** UAT, Customer Validation, Field Trials, Tests to prove compliance with product regulations.

Industrialization

- **SOP (Start Of Production):** Prototype to mass production transition.
- **Operations:** Plant setup (Commissioning) until End of Life.
- **Optimization:** Efficiency & throughput.

Supporting Processes

- **Quality & Compliance:** QA, Configuration Management.
- **Issue Management:** Change Requests, Problem Resolution.
- **Vendors:** Supplier Monitoring & Documentation.

Sales Management

- **Pipeline:** Opportunity Management, Quotation, Pricing.
- **Contracts:** Negotiation & Agreement handling.
- **Feedback:** Customer loop & relationship management.
- **Trade Promotion Management:** planning, execution, ROI evaluation.
- **Demand & Forecasting:** baseline, sell-in, sell-out.

Supply Chain Processes

- **Planning:** Demand & Forecasting for raw materials and/or purchased components from suppliers, Production Planning.
- **Sourcing:** Procurement, Supplier Qualification.
- **Logistics:** Inventory management & distribution.

Industry Process Standards

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Automotive & Motorsport

- Automotive SPICE
- IATF 16949 (QMS)
- ISO 26262 (Functional Safety)
- ISO/SAE 21434 (Cybersecurity)
- ISO/PAS 21448 (SOTIF)

Manufacturing & Industrial Equipment

- ISO 9001 (QMS)
- ISA-95 (Industrial Automation)
- Industry 4.0 Frameworks
- IEC 62061 (Functional Safety)
- IEC 60204 (Safety of Machinery)
- IEC 13849 (Safety of Machinery)

Railway & Transportation

- EN 50126 (RAMS)
- EN 50128 (Functional Safety - Software)
- EN 50129 (Functional Safety - Signalling)
- ISO/IEC 15288 (Systems Engineering)
- ISO/IEC 12207 (Software Engineering)
- ISO/IEC 24748 (Systems and SW Engineering)

Electrical Machines

- IEC 60034 (Rotating Electrical Machines)
- IEC 61800 (Electrical Drive Systems)
- IEEE standard 1012 (Verification & Validation)
- ISO/IEC 29148 (Requirements Engineering)
- ISO 10007 (Configuration Management)
- ISO 31000 (Risk Management)

Power Electronics

- IEC 62477 (Safety for Power Electronics)
- IPC-2221 (PCB Design)
- IEC 61000 (EMC)
- IEC 60812 (FMEA/FMECA)
- IEC 61025 (FTA)

Aerospace & Defense

- AS9100 (QMS)
- DO-178C (Functional Safety - Software)
- DO-254 (Functional Safety - Hardware)
- MIL-STD-498 (Military Software)
- MIL-STD-882 (Functional Safety - Defense)
- MIL-STD-961 (Defense Specifications)

Life Sciences

- GxP (GMP, GLP, GCP, GDP and GSP)
- ISO 13485 (QMS - Medical Devices)
- IEC 60601 (Medical Devices - Safety)
- IEC 62304 (Medical Devices - Software)
- Title 21 CFR Part 11 (Electronic Signatures)

FMCG (Fast Moving Consumer Goods)

- ISO 22000 (Food Safety)
- HACCP (Food Safety)
- IFS Standards (FMCG Safety and Quality)
- BRC Global Standards (FMCG Safety and Quality)

Fashion & Beauty

- ISO 22716 (Cosmetics GMP)
- CE Regulation 1223/2009 (Cosmetics)
- GPSR 2023/988 (Clothes - Safety)
- ESPR UE Regulation 2024/1781 (Sustainability)
- ISO 20400 (Sustainable Procurement)

Oil & Gas

- API Standards (Spec Q1, Spec Q2, 18LCM)
- ISO 29001 (QMS)
- IEC 61511 (Functional Safety)
- ISO 17776 (Offshore Installations)
- ISO 10418 (Safety for Offshore Installations)
- ISO 15926 (Production Plants)
- EN 50271 and EN 50402 (Gas Detection Systems)

Energy & Metals

- IEC 61508 (Functional Safety)
- ISO 50001 (Energy Management Systems)
- IEC 62890 (LCM for Systems & Components)
- IEC 61850 (Power Utility Automation)
- IEC 19433 (QMS Nuclear Energy Sector)
- IEC 60880 (Nuclear Power Plants - Software)

Infrastructure & Construction

- ISO 19650 (Buildings and Civil Engineering Works)
- ISO 55001 (Asset Management)
- ISO 29481 (Building Information Models)
- ISO 12006 (Building Construction)
- ISO 21502 (Project Management)
- ISO 21511 (WBS and Program Management)
- ISO 15686 (Buildings Service Life Planning)

What Is a Use Case?

1. Concept & Identity

- **Definition:** A specific scenario where AI is applied to transform a process step.
- **Unique ID:** A distinct identifier (e.g. UC_001) for tracking.
- **Short Name:** A concise and descriptive title for the specific application.

2. Data Flow

- **Inputs:** Characteristics of data used as inputs (e.g. files, formats, structure of information, technical meaning of information, etc.).
- **Outputs:** The generated results (e.g. files, formats, structure of information, technical meaning of information, etc.)
- **Data Quality:** Consistency of data formats and contents for inputs and outputs.

3. Process Transformation

- **AS IS (Current):** How engineers currently perform the task (often manual/slow).
- **TO BE (Future):** How ML/AI/GenAI solution performs the task on the inputs.
- **Gap Analysis:** Highlighting the operational shift from human-led to AI-assisted.

4. Value Proposition

- **Added Value:** Clear description of benefits (e.g. 40% effort saving, error reduction, etc.).
- **Business Impact:** Qualitative or quantitative link to business goals.
- **KPIs:** Metrics used to measure success (Time, Cost, Quality).

Make vs Buy Analysis

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MAKE (Internal)

Developing custom solutions in-house using internal engineering resources and specific skill sets.

Why Choose Make?

- ✓ **Strategic Differentiation:** Builds unique Intellectual Property (IP) that competitors don't have.
- ✓ **Total Control:** Full ownership of data, roadmap and security.
- ✓ **Tight Integration:** Seamless fit with legacy internal systems.
- ✓ **Gap Filling:** No viable commercial tool exists on the market.

BUY (Market)

Adopting existing commercial solutions (SaaS, COTS) available from vendors in the market.

Why Choose Buy?

- ✓ **Speed to Value:** Immediate implementation vs months of development time.
- ✓ **Lower Risk:** Proven technology with established support.
- ✓ **Best Practices:** Vendor updates bring industry innovations automatically.
- ✓ **Commodity:** Why reinvent the wheel for standard processes?

Decision Criteria:

- Strategic Fit
- Time-to-Market
- Total Cost of Ownership
- IP Sensitivity
- Talent Availability
- Scalability

The Challenge Goal

Each Team is asked to demonstrate analytical skills by selecting a balanced portfolio of Use Cases to be presented to Top Management within an Executive Proposal

Phase 1: Discovery & Definition

1 Select Industry

Choose a target industry sector that aligns with your team's background and interests.

2 Analyze Needs

Identify needs or pain points specific to companies within your chosen industry sector.

3 Focus Process Areas

Select some specific process areas you would like to concentrate on. Suggestion: choose process areas you know better because maybe you experienced in university lab activities and/or practical experiences in order to let your technical awareness to fully emerge.

4 Define Use Cases

Brainstorm and document as many AI/ML/GenAI Use Cases as possible using the assigned template.

5 Map to Process Standards

Associate each defined Use Case with a specific process area from the relevant industry standard.

Phase 2: Analysis & Proposal

6 Market Tool Analysis

Research existing market tools. Collect vendor data on features, performance, KPIs and benefits.

7 Define Clustering Criteria

Establish at least 2 criteria (e.g., Feasibility, Impact, Innovation, etc.) to categorize your Use Cases. Important: a detailed definition of criteria used with related scoring strategies is fundamental to add value to your outcomes.

8 Prioritize Use Cases

Rank and prioritize your identified Use Cases based on the criteria you defined.

9 Prepare Executive Proposal

Select **2 Make** (in-house development) + **2 Buy** (market tool purchase) Use Cases. Justify choices and estimate business benefits.

10 Leverage GenAI and Personal Networking

The deep usage of GenAI tools (e.g. ChatGPT, Claude, Gemini, Perplexity, Grok, etc.) for working on the outcomes is allowed but all used tools shall be clearly mentioned in the final presentation explicitly indicating which tool has been used for doing what). External outreach is allowed and personal networking can be fully exploited without any issues.

Final Deliverables Structure

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Part 1
Team & Context

- **Team Intro:** Names, Surnames and academic backgrounds.
- **Industry:** Reference industry sector selected with motivation.

Part 2
Analysis & Scope

- **Needs & Process:** Identified pain points and process areas.
- **AI Tools:** List of GenAI tools used.
- **External Contacts:** List of networking contacts, job roles and insights gathered.

Part 3
Use Case Details

- **Use Case Tables:** Standard table template for each Use Case.
- **Core Data:** ID, Short Name, Process Area, Inputs, Outputs, AS IS, Current Criticalities, TO BE, Added Value.
- **Market Analysis:** Tool Name, Vendor, Website, Features, Performances, KPIs.

Part 4
Strategy

- **Clustering:** Define clear clustering criteria & motivate choices.
- **Scoring:** Evaluate each Use Case against defined criteria.
- **Prioritization:** Strategy definition based on scoring results.

Part 5
Executive Proposal

- **Final Selection:** 2 Make (in-house development) + 2 Buy (market tool purchase) Use Cases.
- **Justification:** Why these specific choices?
- **Benefits:** Qualitative and quantitative estimation of added value & business impact.

Submission Package

D1 - Presentation (PPTX)
Contains Parts 1, 2, 3, 4, 5. Full details.

D2 - Presentation (PDF)
Contains Parts 1, 2, 3, 4, 5. Identical to D1.

D3 - Video Pitch (MP4)
Contains at least Parts 1, 2, 5. Max 10 mins.

Video Deliverable Requirements

Technical Specifications

- **Format:** Standard .mp4 file.
- **Duration:** Strictly maximum 10 minutes.
- **Naming:** Team_Number.mp4 (e.g. Team_05.mp4).
- **Audio:** Ensure clear narration in English language split among all team members.

Required Contents

- **Team Intro:** Briefly introduce all team members and backgrounds.
- **Context:** Explain chosen industry sector, focused process areas and identified needs.
- **Executive Proposal:** Present your Final Selection of Use Cases [2 "Make" (in-house development) + 2 "Buy" (market tool purchase)].
- **Justification:** Explain the "Why" behind your choices.

Strategic Goal

- **Convince Top Management:** Pitch for investment approval.
- **Show Business Value:** Quantify potential savings or gains.
- **Demonstrate Competence:** Show your technical and communication skills.

Presentation Tips

- **Storytelling:** Create a logical flow from problems to solutions.
- **Visuals:** Show your tables and/or matrices (don't just read text).
- **Clarity:** Be concise, professional and persuasive.
- **Audio Quality:** Ensure all speakers are audible.

Evaluation Criteria & Next Steps

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Evaluation Scoring Breakdown

Total: 100%

Use Case Definition Quantity and technical consistency of identified Use Cases.	25%
Process Mapping Quality of mapping Use Cases to standard process areas.	5%
Market Tools Analysis Capability analysis and quality mapping of market tools.	25%
Prioritization Approach Criteria definitions and logic used for Use Case prioritization.	10%
Executive Proposal Quality, persuasion and technical consistency of the proposal.	35%

After the Hackathon - Next Steps

What happens next for the best teams?

Final Defense

Top 8 Teams will be invited to NTT DATA Torino headquarters for a half-day session. You will present your work and answer questions from our process experts.

Awarding Ceremony

At the end of the defense day, the **Top 3 Teams** will be proclaimed winners and awarded prizes for their outstanding performance.

Use Case Definition Template

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Use Case ID	Short Name	Reference Process Area	Characteristics of Inputs	Characteristics of Outputs
UC_001	*{e.g., Automated Req. Parsing}*	*{e.g., Requirements Management}*	*{Data format, volume, frequency...}*	*{Report structure, API response...}*

AS IS Situation

Current Process Description

{Describe how engineers currently perform this task manually step-by-step...}

Current Process Criticalities and Pain Points

{High effort, error-prone, lack of traceability, bottlenecks...}

TO BE Situation (AI Enabled)

AI/GenAI Application Description

{Explain how the AI model processes inputs to generate outputs automatically...}

Expected Added Value & Benefits

{Quantifiable benefits: -30% effort, +20% quality, cost reduction estimates...}

Commercial Tools Market Analysis

List at least 2 potential market solutions

Tool Name	Tool Vendor	Reference Website	Tool Features	Expected Performances & KPIs
{Name}	*{Vendor}*	*{URL}*	*{Key features list}*	*{Accuracy, Speed...}*
Tool Name	**Tool Vendor**	**Reference Website**	**Tool Features**	**Expected Performances & KPIs**
{Name}	*{Vendor}*	*{URL}*	*{Key features list}*	*{Accuracy, Speed...}*

Good Luck
& Have Fun!

We look forward to seeing your innovative solutions.

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